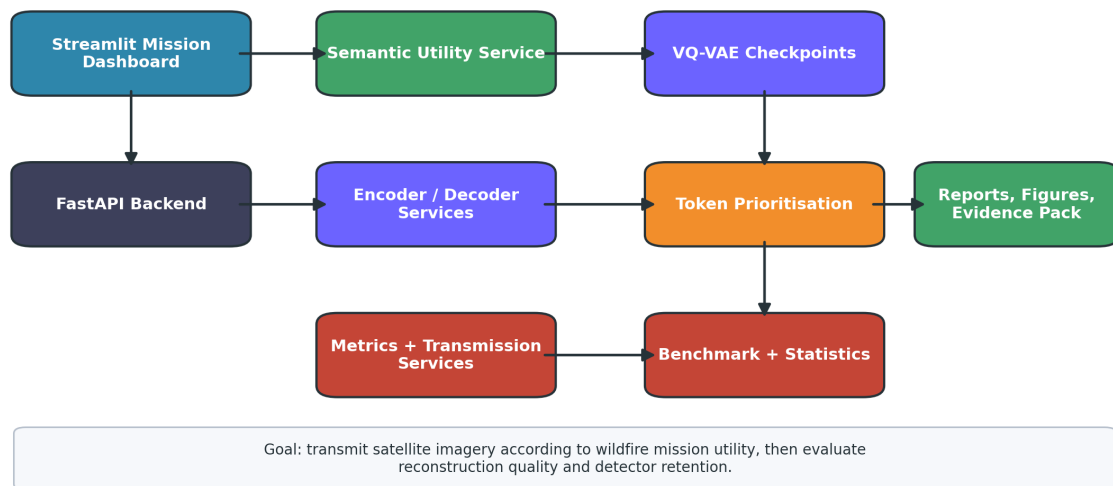


Founder and Visa Evidence Package

Onboard AI Software for Mission-Aware Satellite Downlink Optimisation

CompressAI System Architecture



Purpose: a structured evidence package for Innovator Founder preparation, Global Talent evidence planning, incubator outreach, and space-tech discussions.

1. Executive Business Summary

The proposed venture develops onboard AI software that reduces satellite downlink bandwidth while preserving mission-critical Earth Observation information. The first mission focus is wildfire monitoring, where preserving fire, smoke, burn-scar, and affected-terrain evidence matters more than reconstructing every background pixel perfectly.

Positioning statement

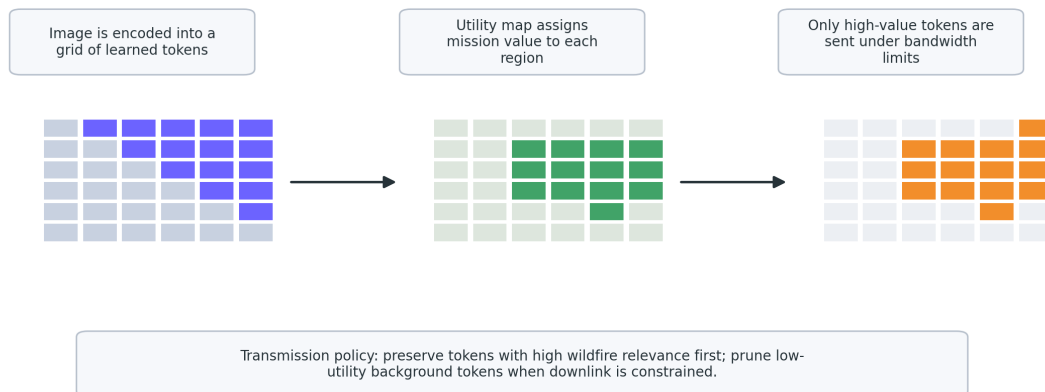
Mission-aware semantic communication software for resource-constrained Earth Observation satellites.

2. Problem and Customer Pain

- Small satellites and CubeSats face limited downlink capacity, power budgets, and contact windows.
- Emergency monitoring needs rapid transmission of useful information, not necessarily full-resolution imagery.
- Conventional compression optimises visual fidelity, but mission teams care about whether operational information survives.
- Wildfire monitoring requires prioritising active fire, smoke, burn scars, and affected terrain.

3. Product Concept

Token-Based Semantic Transmission

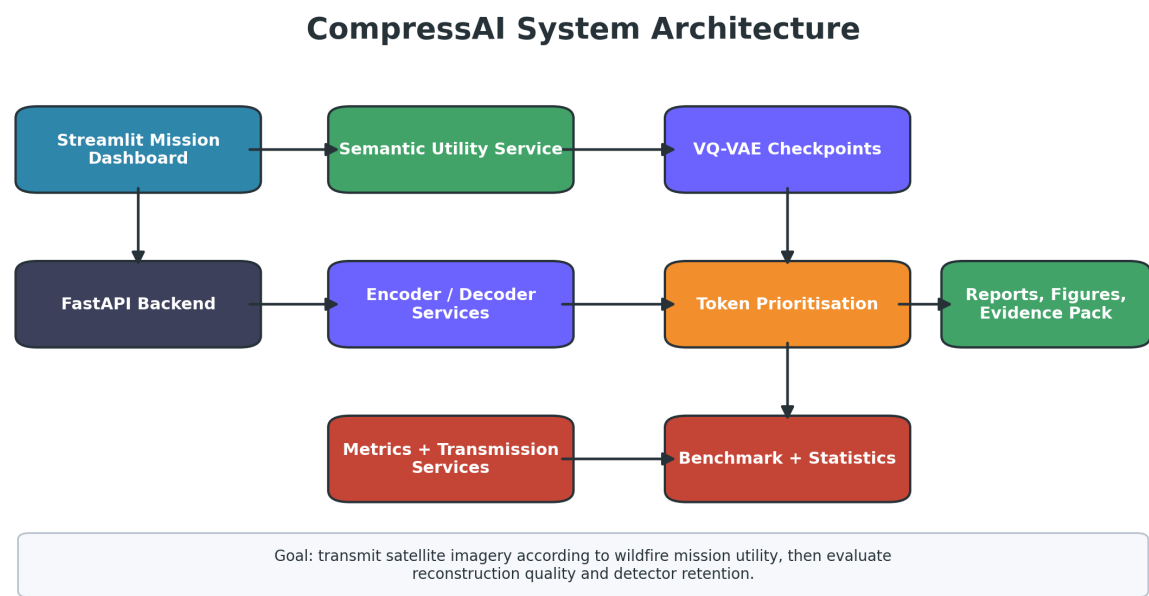


The product is a software layer that sits between onboard sensing and downlink. It analyses image utility, encodes imagery into learned tokens, prioritises mission-relevant tokens, and transmits a smaller payload that preserves operational value.

4. Innovative, Viable, Scalable

Criterion	Evidence
Innovative	The system is not generic compression; it prioritises learned satellite-image tokens according to mission utility and communication constraints.
Viable	The repository contains a working backend, frontend, VQ-VAE pipeline, benchmark framework, Sentinel-2 validation, diagrams, and reproducible reports.
Scalable	The method can extend from wildfire to flood, maritime, infrastructure, defence, and civil protection missions.
Founder credibility	The project has real code, test coverage, benchmark evidence, and structured application packages.

5. Platform Architecture

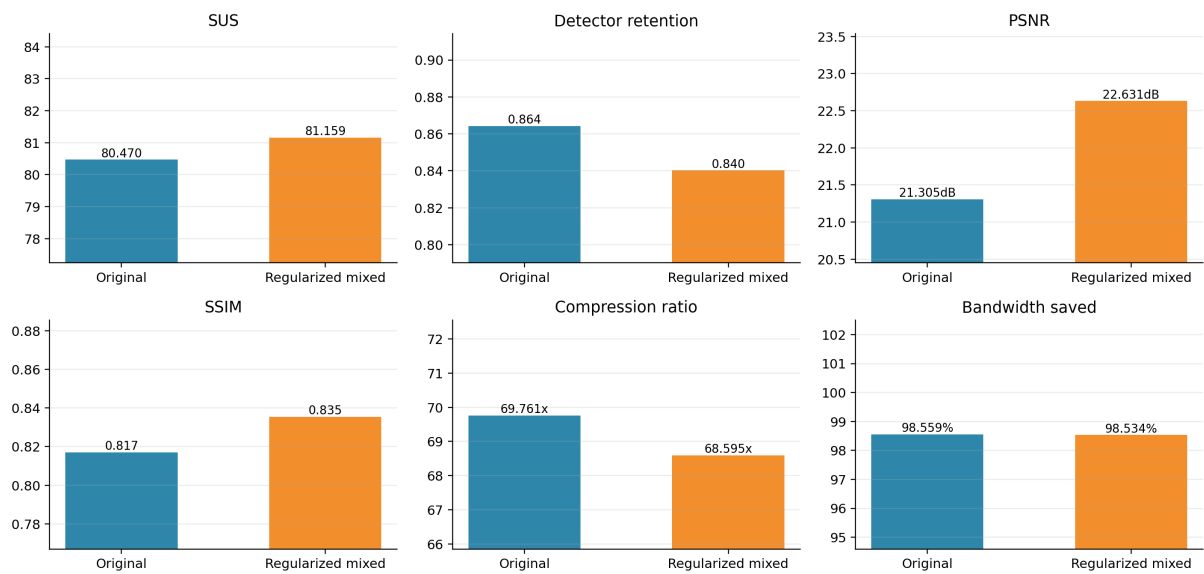


6. Evidence From Sentinel-2 Validation

Metric	Original VQ-VAE	Regularized mixed-domain VQ-VAE	Interpretation
SUS	80.470	81.159	Candidate slightly higher, but not significant enough alone for promotion.
Detector retention	0.8642	0.8403	Original is safer for wildfire mission-utility preservation.
PSNR	21.305	22.631	Candidate improves visual reconstruction by +1.326 dB.
SSIM	0.8170	0.8354	Candidate improves structural similarity.
Compression ratio	69.76x	68.60x	Original remains slightly stronger for compression ratio.
Bandwidth saved	98.56%	98.53%	Both preserve very high communication savings.

7. Communication Efficiency Result

Sentinel-2 500-Patch Benchmark: Original vs Regularized VQ-VAE



The key commercial result is not that the system beats every existing codec. The stronger claim is that it provides a mission-aware framework for deciding what information to preserve when communication is constrained. Current benchmark evidence shows very high bandwidth saving while exposing a measurable trade-off between reconstruction quality and detector utility.

8. Current Product Readiness

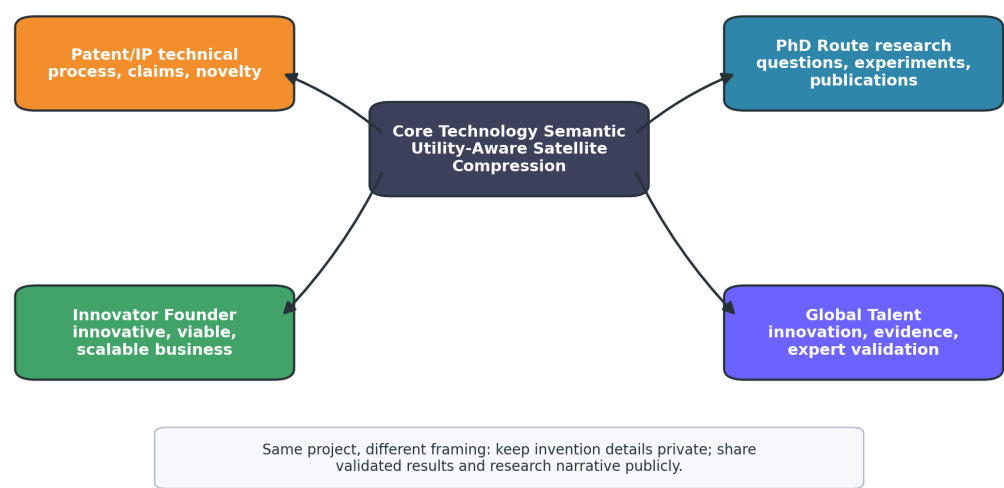
Area	Current status	What is still needed
Prototype	Working API, dashboard, encoder/decoder, token selector, benchmark scripts.	Short demo video and hosted demo walkthrough.
Evidence	Sentinel-2 benchmark, statistical results, diagrams, reports.	More external dataset validation and third-party feedback.
IP	Private invention disclosure and attorney brief drafted.	Patent attorney review before public claim-level disclosure.
Commercial	Founder/visa package prepared.	Customer interviews, letters of interest, incubator feedback.
Deployment	Docker and modular services exist.	Jetson/onboard-class hardware validation.

9. Global Talent Evidence Mapping

Evidence category	Current evidence	Next action
Innovation	Original semantic satellite compression prototype.	Publish a technical preprint and record demo.
Technical contribution	GitHub repository, modular architecture, benchmark framework.	Collect expert reviews from EO and compression researchers.
Research promise	PhD proposal, statistical benchmark, diagrams, supervisor package.	Begin supervisor outreach and workshop submission.
Commercial promise	Founder package, IP brief, space-sector use case.	Collect customer discovery notes and letters of interest.
External validation	Not yet enough.	Request written feedback from supervisors, incubators, and patent attorney.

10. Founder Route Positioning

Evidence Positioning: Patent, PhD, Innovator Founder, Global Talent



11. Target Organisations

Category	Targets
Space incubators	ESA BIC, Satellite Applications Catapult, UK space accelerators, Luxembourg space ecosystem.
Research groups	University of Luxembourg SnT, TU Delft, DLR, ESA Phi-Lab aligned groups, UK EO and onboard AI labs.
Early users	CubeSat teams, EO analytics companies, wildfire monitoring teams, environmental monitoring groups.
Advisors	Patent attorney, EO scientist, satellite systems engineer, startup mentor.

12. 60-Day Action Plan

- Record a 2 to 3 minute demo video showing image input, utility heatmap, token selection, reconstruction, and metrics.
- Send supervisor package to 20 targeted PhD supervisors.
- Contact 5 to 8 incubators or space programmes with the founder package.
- Book patent attorney consultation using the private brief.
- Prepare a preprint or workshop submission using the current Sentinel-2 evidence.

- Collect written feedback and letters of interest for visa or founder evidence.

This package is evidence framing and not immigration advice. It is intended to support conversations with endorsing bodies, incubators, supervisors, and legal professionals.